

Adaptive Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

Aditya Dendukuri

University of California, Santa Barbara

A probability ODE for stochastic chemical kinetics

Setting: N species reacting in a well-mixed vessel. State $\mathbf{n} \in \mathbb{Z}_{\geq 0}^N$: molecule counts. Reaction r fires at rate $\alpha_r(\mathbf{n})$, shifts state by ν_r .

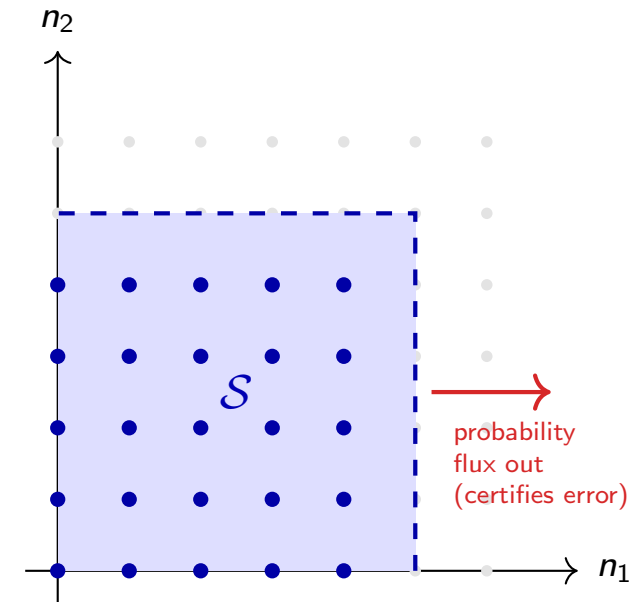
Joint probability $p(\mathbf{n}, t) = \Pr[N(t)=\mathbf{n}]$ — all species counts simultaneously — satisfies the **Chemical Master Equation (CME)**:

$$\frac{dp}{dt} = A p, \quad A_{\mathbf{n}+\nu_r, \mathbf{n}} = \alpha_r(\mathbf{n}).$$

Finite State Projection [Munsky & Khammash 2006]

Truncate to $\mathcal{S} \subset \mathbb{Z}_{\geq 0}^N$; solve $\dot{p}_{\mathcal{S}} = A_{\mathcal{S}} p_{\mathcal{S}}$.
Error = probability flux leaving $\mathcal{S}$ — bounded at runtime.
Expand $\mathcal{S}$ only when flux exceeds tolerance.

Challenge: $|\mathcal{S}|$ grows combinatorially. Coupling K voxels makes this $n_{\max}^K$ — dramatically worse.



State space truncated to $\mathcal{S}$.

The RDME: two levels of state

Level 1 — within each voxel k :

$$\mathbf{n}_k = (n_k^1, \dots, n_k^N) \in \mathbb{Z}_{\geq 0}^N$$

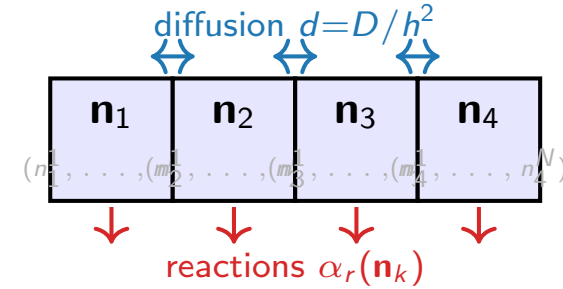
n_k^s = count of species s in voxel k . Same CME as the well-mixed case, one instance per voxel.

Level 2 — across all K voxels:

$$(\mathbf{n}_1, \dots, \mathbf{n}_K) \in (\mathbb{Z}_{\geq 0}^N)^K$$

$p(\mathbf{n}_1, \dots, \mathbf{n}_K, t)$ = probability over the full configuration; $\dot{p} = Ap$.

- **Reactions in voxel k :** change $\mathbf{n}_k$ only
- **Diffusion $k \leftrightarrow k+1$:** molecule hops — changes both $\mathbf{n}_k$ and $\mathbf{n}_{k+1}$



$\mathbf{n}_k$ is the state of voxel k — a vector, not a species label.

- $N=1$: $\mathbf{n}_k = n_k$ (scalar)
- $N>1$: $\mathbf{n}_k \in \mathbb{Z}_{\geq 0}^N$; each voxel adds N dimensions

The full state space is the *product* of K voxel state spaces.

State space: two scales

Per voxel: $n_{\max}^N$ states.

K voxels: $n_{\max}^{NK}$ states total.

The RDME generator: reactions on diagonal, diffusion off-diagonal

	$\mathbf{n}_1$	$\mathbf{n}_2$	$\mathbf{n}_3$	$\mathbf{n}_4$
$\mathbf{n}_1$	A_1	D_{21}		
$\mathbf{n}_2$	D_{12}	A_2	D_{32}	
$\mathbf{n}_3$		D_{23}	A_3	D_{43}
$\mathbf{n}_4$			D_{34}	A_4

$N=1$ shown. For $N>1$: each block is $n_{\max}^N \times n_{\max}^N$. Matrix size: $n_{\max}^{NK} \times n_{\max}^{NK}$.

Each axis = one voxel state $\mathbf{n}_k$.

- A_k : reaction generator for voxel k — diagonal block; changes $\mathbf{n}_k$ only
- $D_{k,k+1}$: diffusion coupling — off-diagonal, nearest-neighbor; changes $\mathbf{n}_k$ and $\mathbf{n}_{k+1}$

$A = A_{\text{rxn}} + A_{\text{diff}}$, block-tridiagonal.

Sparse: each transition changes at most two neighboring voxel states.

The challenge

Size $n_{\max}^{NK}$ explodes as K grows. **Goal:** reduce without losing accuracy.

Coarsening a voxel pair: restriction and Binomial split-back

Pair (n_1, n_2) : diffusion conserves the total $\bar{n} = n_1 + n_2$. Shown for $S=1$, $m=2$; extends to any S , m by independence of species.

Marginalize $(\mathcal{R})$ — exact

$$p^c((\bar{n}_s)_s) = \sum_{\substack{\text{configs} \\ \text{with totals } (\bar{n}_s)}} p(\dots).$$

Split back $(\mathcal{P})$ — independent per species

$$\mathcal{P} = \prod_{s=1}^S \text{Bin}(\bar{n}_s, \frac{1}{m}).$$

($m=2$: Binomial; $m>2$: Multinomial.)
Valid when diffusion dominates within the group.

State-space reduction

Per group: $n_{\max}^{mS} \rightarrow n_{\max}^S$.

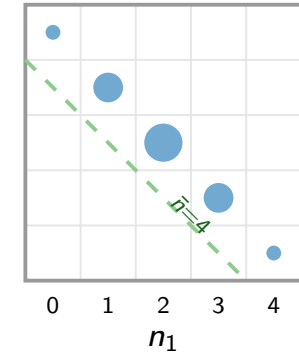
K/m groups all coarse: $n_{\max}^{KS} \rightarrow n_{\max}^{KS/m}$.

$\mathcal{R}$
sum along
diagonals

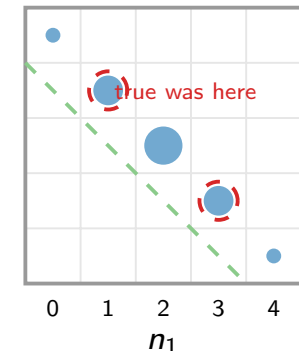
same $\bar{n}$ for
both cases!

$\mathcal{P}$
 $\text{Bin}(\bar{n}, \frac{1}{2})$
splits back

Uniform — $\mathcal{P}(\mathcal{R} p) \approx p$. ✓

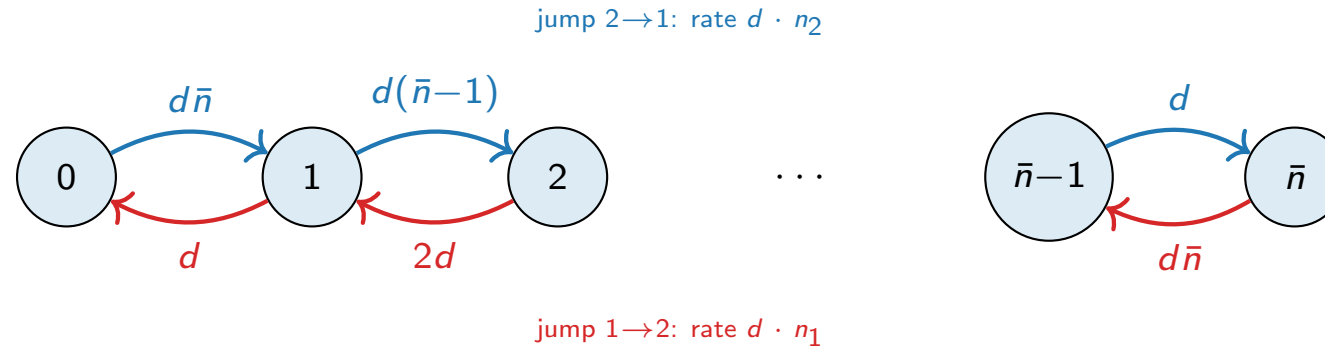


Front — Binomial peaks at centre;
true is bimodal. ✗



Why the Binomial: diffusion alone equilibrates to $\text{Bin}(\bar{n}, \frac{1}{2})$

Fix total $\bar{n} = n_1 + n_2$. Within-group diffusion alone is a birth-death chain on $\{0, \dots, \bar{n}\}$:



Detailed balance:

$$\pi(n_1) \cdot d(\bar{n} - n_1) = \pi(n_1 + 1) \cdot d(n_1 + 1)$$

$$\frac{\pi(n_1 + 1)}{\pi(n_1)} = \frac{\bar{n} - n_1}{n_1 + 1} = \frac{\binom{\bar{n}}{n_1 + 1}}{\binom{\bar{n}}{n_1}}$$

$$\boxed{\pi(n_1) = \text{Bin}(\bar{n}, \frac{1}{2})} \quad \text{(exact)}$$

Exact vs. approximate

Exact: $\text{Bin}(\bar{n}, \frac{1}{2})$ is the stationary distribution for any $d > 0$.

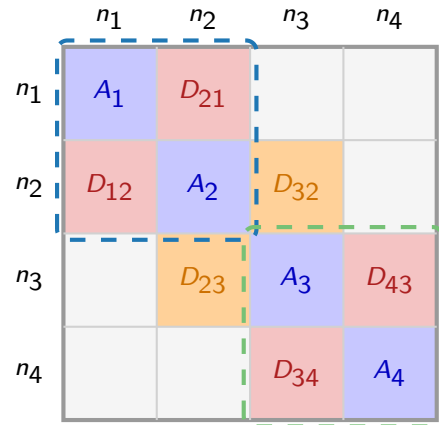
Approximate: valid when $d_s \gg \alpha_r$ for all species s .

$S > 1$: one independent chain per species; joint prolongation = $\prod_s \text{Bin}(\bar{n}_s, 1/m)$.

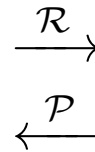
$m > 2$: $\text{Bin} \rightarrow \text{Mult}(\bar{n}_s; 1/m, \dots, 1/m)$.

What grouping does to the generator

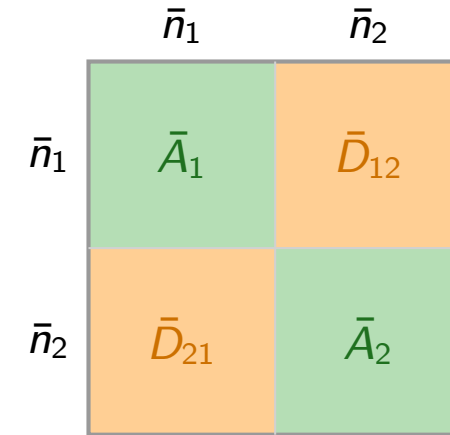
Full-resolution generator A — 4 voxels:



State space: $n_{\max}^4$



Coarse generator $\bar{A}$ — 2 groups:



State space: $n_{\max}^2$ ($K=8 \rightarrow 4$: 3×10^4 smaller)

- D_{12}, D_{21} **disappear** — diffusion inside each group is absorbed into the Binomial
- $\bar{A}_j = \mathcal{R}(A_{2j-1} + A_{2j})\mathcal{P}$ — Binomial-averaged reactions for pair j
- $\bar{D}_{23}$ — diffusion at the group boundary (rescaled by $\mathcal{R}, \mathcal{P}$)

Numerical example: Schlögl model ($S=1$, $m=2$, $K=8$)

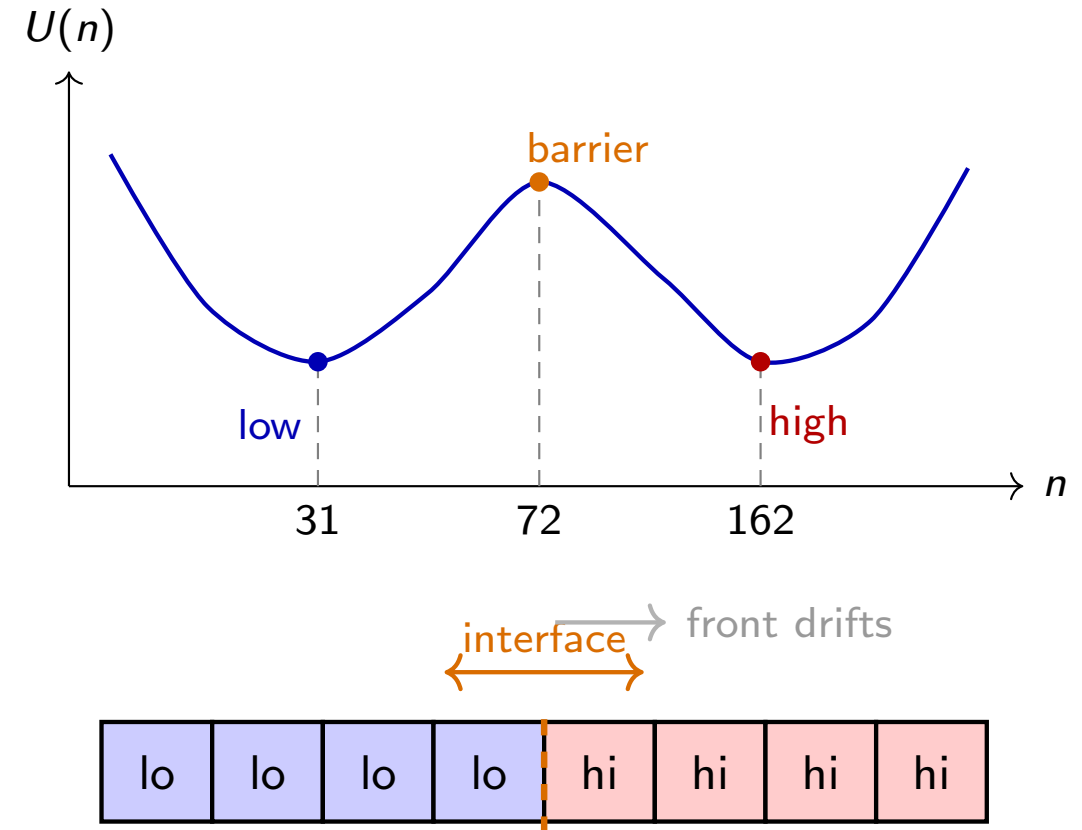
Schlögl model: each voxel has two stable molecule counts.

Single-voxel fixed points

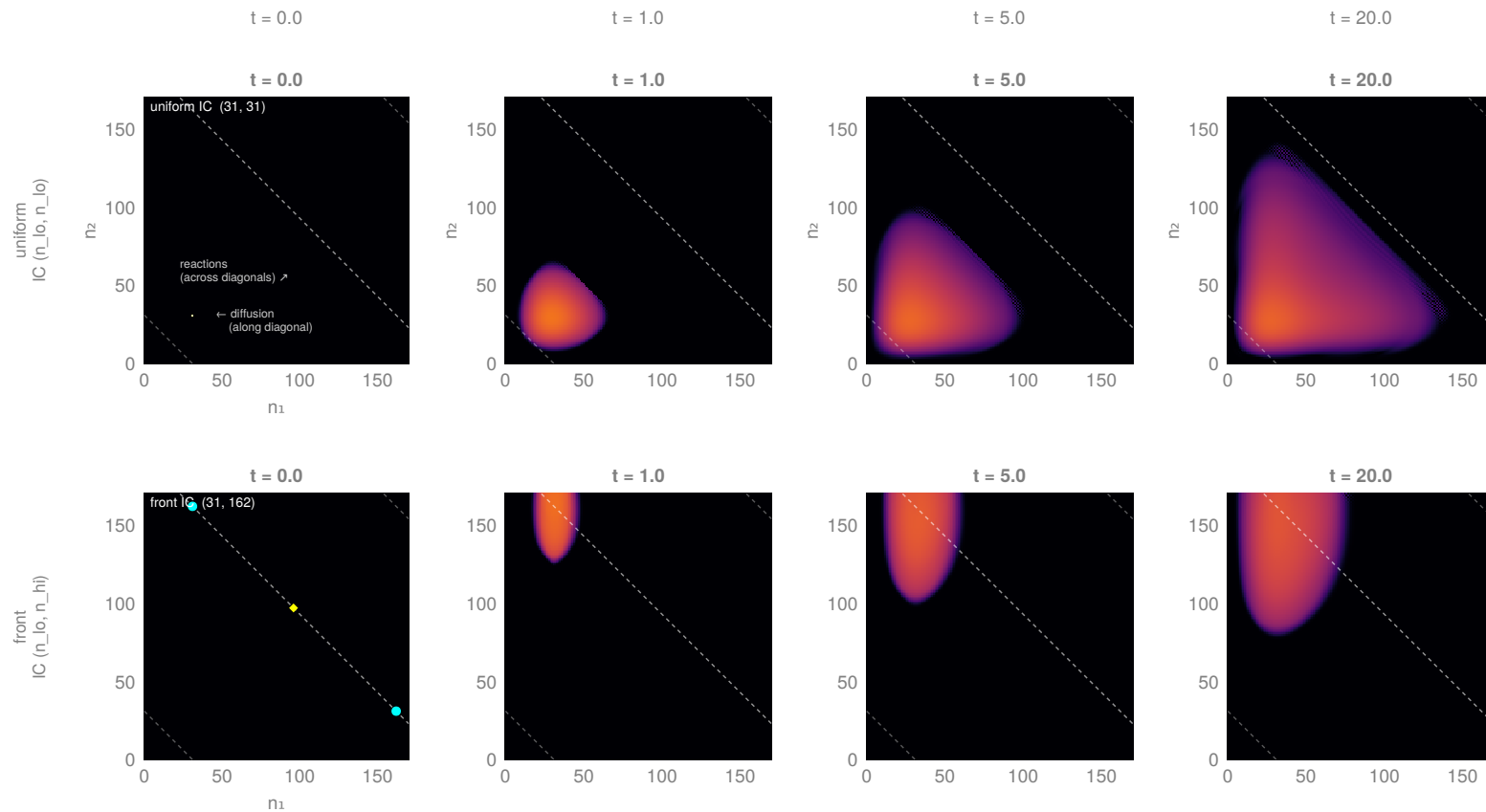
$$n_{\text{low}} = 31, \quad n_{\text{barrier}} = 72, \quad n_{\text{high}} = 162.$$

A **front** forms when neighboring voxels settle in opposite stable states. The interface drifts stochastically along the domain.

Diffusion rate $d=D/h^2=0.08$ — slow relative to bistable reactions, which sustains the front indefinitely.



The Binomial approximation fails at the front



$K=2$ Schlögl ($D = 0.005$, $d = 0.02$). **Top (uniform, (31, 31))**: diffusion moves probability along constant-total lines; conditional $\rightarrow \text{Bin}(\bar{n}, \frac{1}{2})$ — grouping is valid. **Bottom (front, (31, 162))**: bistable reactions outrun diffusion; probability stays at the corners; conditional $\neq \text{Bin}(\bar{n}, \frac{1}{2})$ — grouping is invalid. Dashed diagonals = constant-total lines.

The imbalance criterion α_j

Per-species imbalance within group j :

$$\alpha_{j,s} = \frac{|\mu_{2j-1,s} - \mu_{2j,s}|}{\mu_{2j-1,s} + \mu_{2j,s}}, \quad \alpha_j = \max_s \alpha_{j,s}.$$

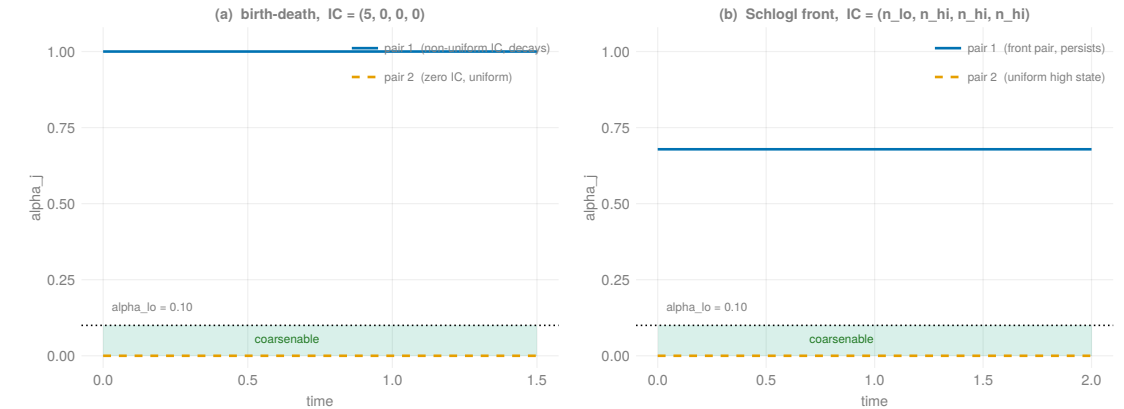
Small α_j : all species balanced — diffusion dominates within the group.

The coarsening rule

Coarsen group j iff $\alpha_j < \alpha_{lo}$.

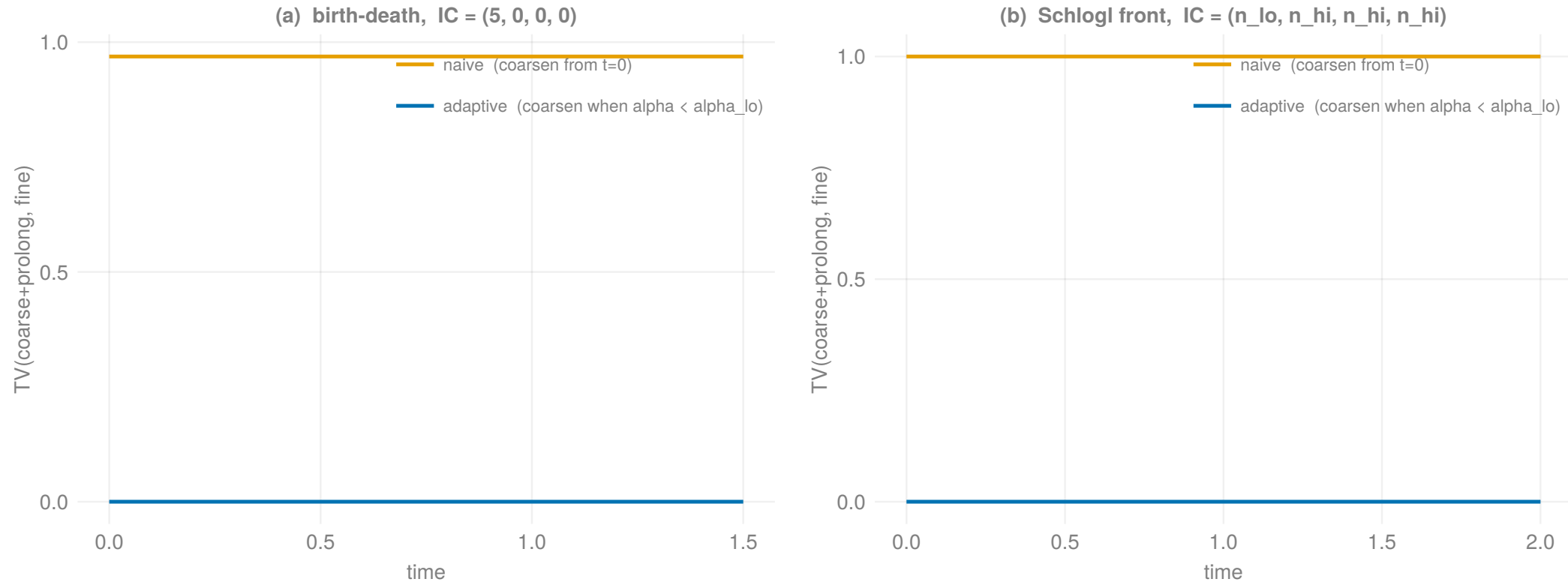
- Non-equilibrium IC: α_j starts large, decays as diffusion acts
- Bistable front: α_j stays large (reactions as fast as diffusion)

Split-back applied only where diffusion dominates for all species.



(a) Birth-death: α_1 decays to zero (diffusion equilibrates the pair). **(b)** Schlögl front: α_1 persists (bistable reactions outrun diffusion). Green band = region where grouping is valid.

Adaptive grouping eliminates the error

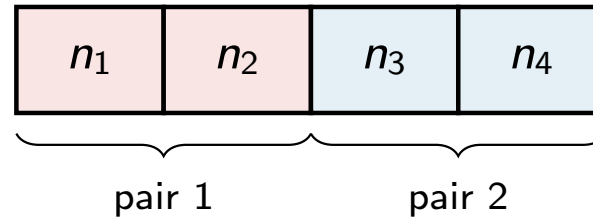


(a) Birth-death: uniform coarsening (Binomial from $t=0$) gives $TV=0.97$ immediately. Adaptive (coarsen only when $\alpha < \alpha_{lo}$): $TV=0$ until equilibration, then small. **(b)** Schlögl front: uniform coarsening gives $TV=1.0$ throughout (orientation destroyed). Adaptive (keep group 1 at fine resolution): $TV=0$ throughout.

The mixed-resolution RDME

Replace (n_{2j-1}, n_{2j}) by the total $\bar{n}_j$ for equilibrated pairs; keep front pairs at full resolution.

F: fine states $\{n_{k,s}\}$ **C: coarse states** $\bar{n}_{j,s} = \sum_{k \in G_j} n_{k,s}$



- **Fine (F):** full joint distribution $\{n_{k,s}\}_{k \in G_j}$; within-group diffusion tracked exactly
- **Coarse (C):** only totals $(\bar{n}_{j,s})_s$; within-group diffusion absorbed into $\prod_s \text{Bin}(\bar{n}_{j,s}, \frac{1}{m})$
- **Evolve directly:** $\dot{p}^M = A^M p^M$ — no convert-back step

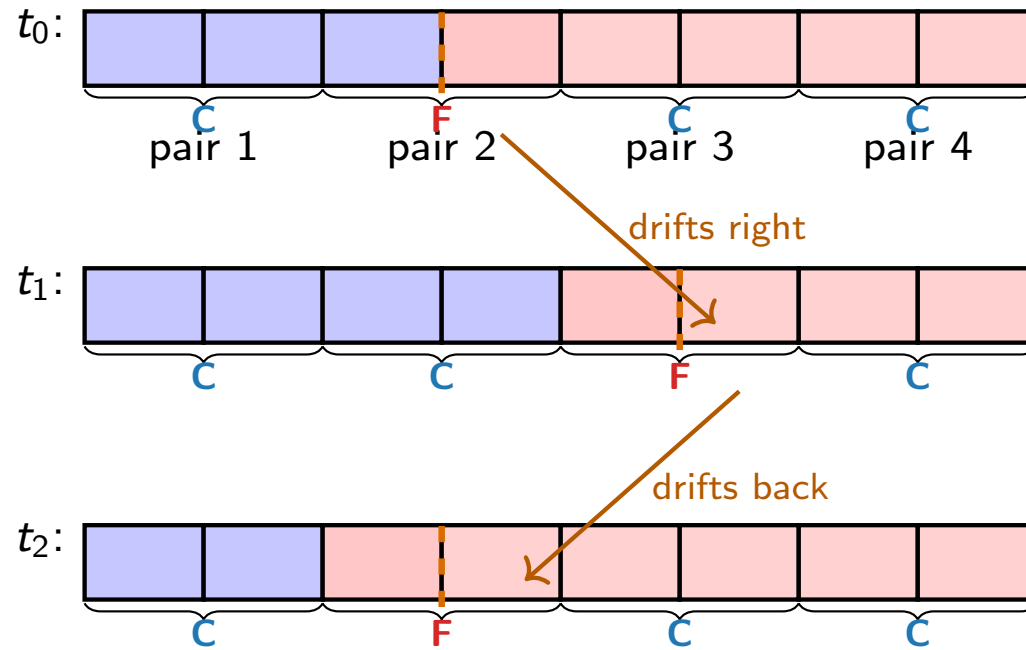
State-space savings

Per group: $n_{\max}^{mS}$ fine $\rightarrow n_{\max}^S$ coarse.

K/m groups all coarse: $n_{\max}^{KS} \rightarrow n_{\max}^{KS/m}$.

Coarsen where $\alpha_j < \alpha_{10}$; keep fine near the front.

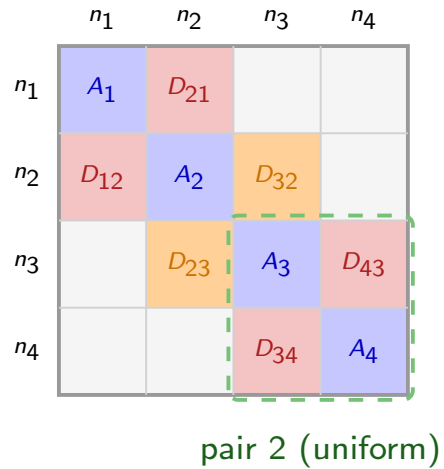
Tracking the drifting front



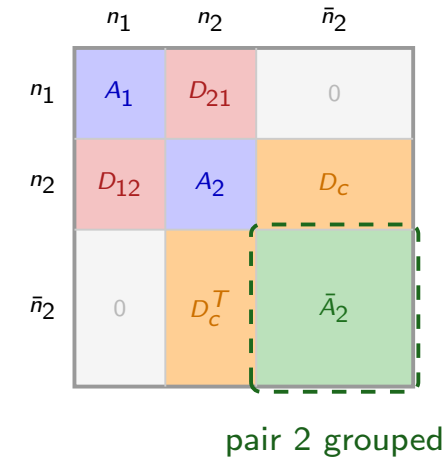
The fine window (**F**) keeps the front pair at full resolution. Groups far from the front (**C** = coarse) stay coarse. Resolution is updated only when the front crosses a pair boundary.

The mixed-resolution generator

Full-resolution generator A ($K = 4$):



Mixed generator A^M (pair 2 grouped):



$\bar{A}_2$: Binomial-averaged reactions — valid since pair 2 is uniform.
 D_c : fine/grouped diffusion coupling: rate $d \cdot n_2$ (fine→grouped),
 $d \cdot \bar{n}_2/2$ (grouped→fine).

Evolving directly on the mixed state

Evolve p^M directly — no split-back step

- 1 **Evolve:** $\dot{p}^M = A^M p^M$ — p^M advances directly.
- 2 **Monitor front:** track mean interface position $\bar{j}$ every step (cheap).
- 3 **Reassign only when front moves:** when $|\bar{j} - \bar{j}_{\text{prev}}| \geq \frac{1}{2}$, coarsen newly uniform groups; refine newly fine groups.

Why not split back after each step?

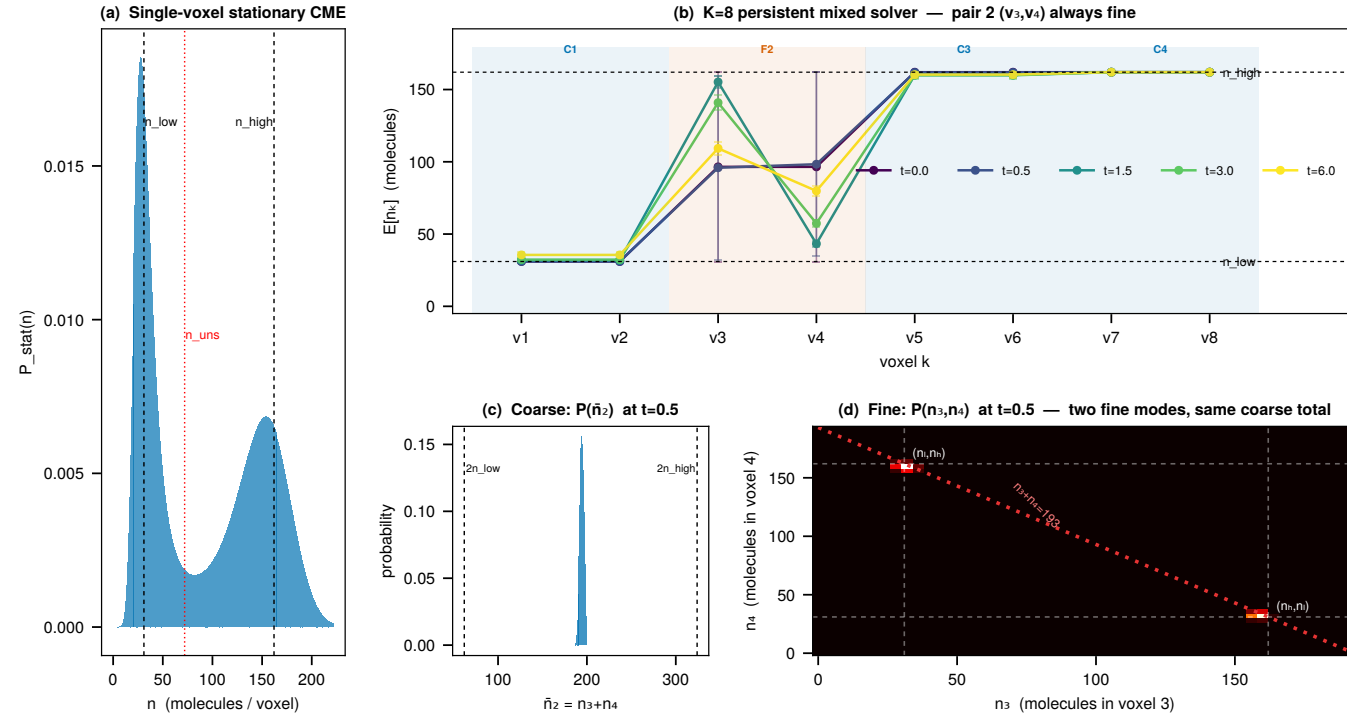
Splitting back forces Binomial at *all* pairs — wrong at the front. Evolving p^M directly avoids this.

Cost: grid reassigned $O(K/2)$ times total (once per group boundary crossed), not $O(n_{\text{steps}})$.

State space: adaptive FSP — expand a pair's sub-space only when flux out exceeds tolerance.

Mixed resolution resolves interface orientation

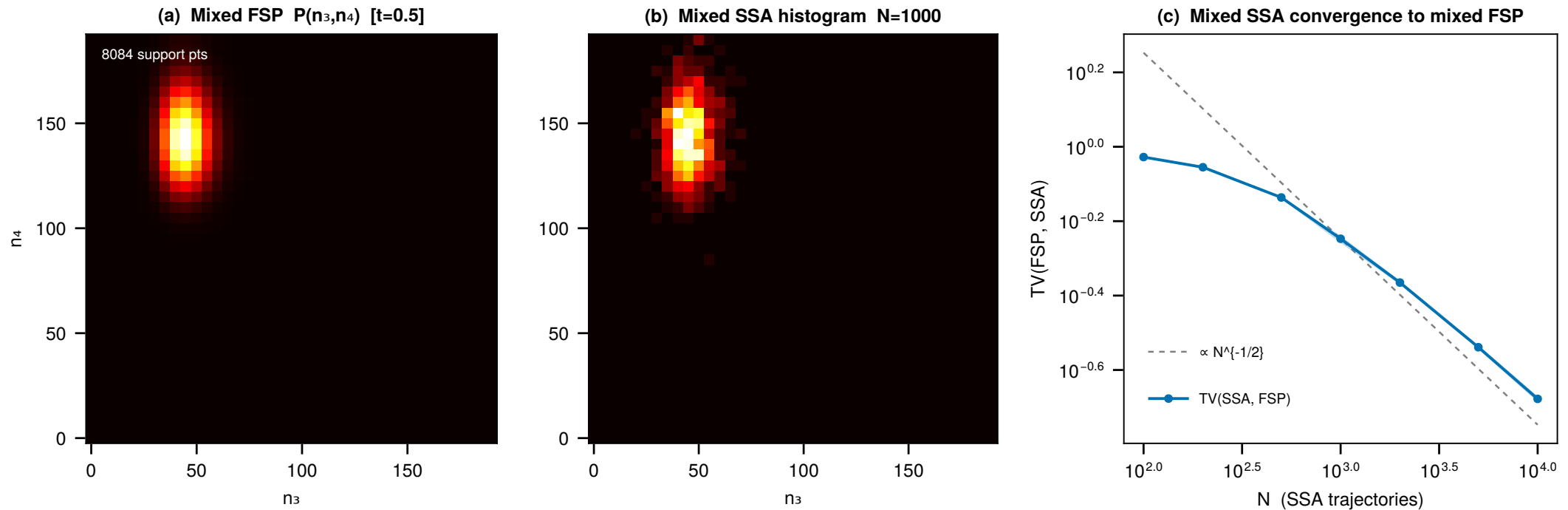
Question: can the mixed state distinguish (n_{lo}, n_{hi}) from (n_{hi}, n_{lo}) , which both map to the same grouped state?



IC: interface neighbors 2 as equal superposition of (n_l, n_h) and (n_h, n_l) . **(c)** Coarse $P(\bar{n}_2)$: single peak at 193 — orientation is lost. **(d)** Fine $P(n_3, n_4)$: two modes along $n_3 + n_4 = 193$ — the mixed state retains the within-pair structure. Full $K=8$ fine solve: intractable ($> 258k$ states).

Mixed generator is self-consistent

Question: does the fine/grouped coupling in A^M introduce systematic error?



Direct ODE solve on mixed state $(\bar{n}_1, n_3, n_4)$ vs. Monte Carlo simulation on the same generator. Means agree to $<1\%$; $TV \sim N^{-1/2}$ (sampling noise only).

Summary

1. The RDME state space is exponential; diffusion enables grouping.

When diffusion between two neighboring voxels is fast relative to reactions, it equilibrates their joint distribution to $\text{Bin}(\bar{n}, \frac{1}{2})$. Grouping = replacing (n_1, n_2) by $\bar{n} = n_1 + n_2$: exact marginalization $\mathcal{R}$, Binomial split-back $\mathcal{P}$. State space: $n_{\max}^K \rightarrow n_{\max}^{K/2}$.

2. Grouping fails at a bistable front.

Reactions keep neighboring voxels in opposite stable states — faster than diffusion can equilibrate them. The imbalance criterion $\alpha_j = |\mu_{2j-1} - \mu_{2j}| / (\mu_{2j-1} + \mu_{2j})$ detects this: large α_j means do not group pair j .

3. Adaptive mixed-resolution evolution.

Keep a fine window around the front (exact within-pair distribution); group the rest (Binomial split-back valid where diffusion dominates). Evolve $\dot{p}^M = A^M p^M$ directly; reassign only when the front crosses a pair boundary.

Open questions: replace α_j with a rigorous diffusion flux-balance bound; multi-species; deeper hierarchies.

A naive approach: restrict-then-prolong at every step

What if we grouped all pairs, solved the grouped ODE, then split back each step?

Naive V-cycle over time step Δt

- ① **Smooth** — apply within-group diffusion until the conditional equilibrates:

$$\tilde{p} = e^{\tau A_{\text{diff, within}}} p.$$

- ② **Marginalize** — sum within pairs:

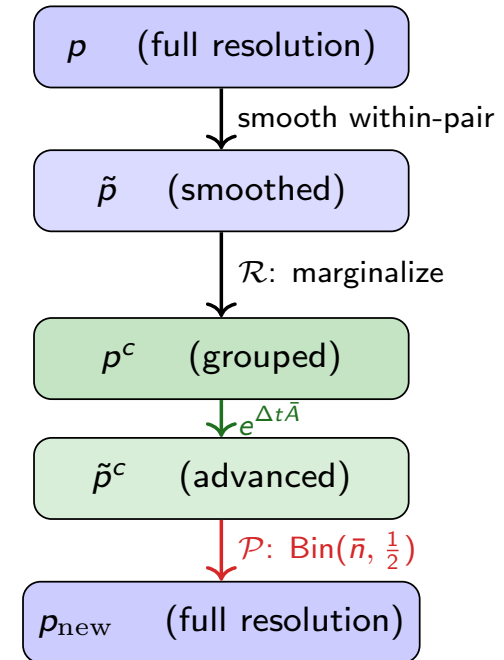
$$p^c = \mathcal{R} \tilde{p}, \quad \mathcal{R}_{\bar{n}, s} = \mathbf{1}[n_1 + n_2 = \bar{n}].$$

- ③ **Grouped solve** — advance the grouped ODE:

$$\tilde{p}^c = e^{\Delta t \bar{A}} p^c.$$

- ④ **Split back** — reconstruct within-pair split:

$$p_{\text{new}} = \mathcal{P} \tilde{p}^c, \quad \mathcal{P}_{s, \bar{n}} = \text{Bin}(\bar{n}, \frac{1}{2}).$$



Step 4 fails at the front

Binomial split-back applies to *all* pairs — including the front, where the conditional is not Binomial. **Solution:** evolve the mixed state p^M directly (no split-back step at all).

Interface detection indicators

Within-pair imbalance: front inside pair j

$$\alpha_j = \frac{|\mu_{2j-1} - \mu_{2j}|}{\mu_{2j-1} + \mu_{2j}}$$

Large when the two voxels in pair j are imbalanced — diffusion has not equilibrated them.

Between-pair imbalance: front at a pair boundary

$$\beta_j = \frac{|\mu_{2j} - \mu_{2j+1}|}{\mu_{2j} + \mu_{2j+1}}$$

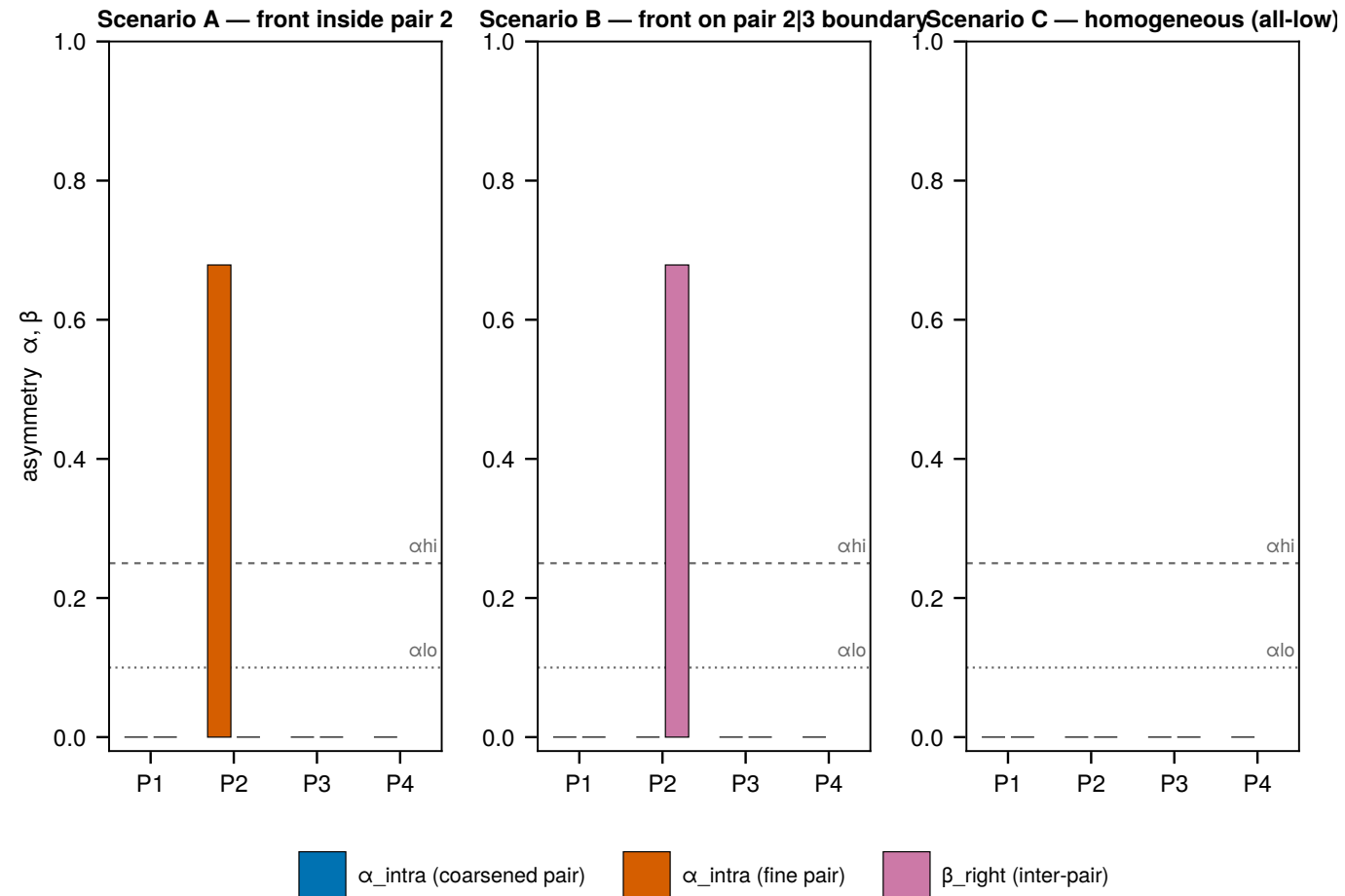
Large when voxels $2j$ and $2j+1$ (straddling a pair boundary) are imbalanced.

Effective indicator and grouping rule

$$\alpha_j^{\text{eff}} = \max(\alpha_j, \beta_{j-1}, \beta_j).$$

Flag group j fine if $\alpha_j^{\text{eff}} > \alpha_{\text{hi}}$. Coarsen group j only if $\alpha_j^{\text{eff}} < \alpha_{\text{lo}}$ (hysteresis).

Representative configurations



A: front interior to pair 2, detected by α_2 . **B:** front on the pair 2|3 boundary, detected by β_2 . **C:** uniform — all pairs grouped.

- Erban, Chapman, Maini (2007). Stochastic simulations of reaction-diffusion. *arXiv:0704.1908*.
- Munsky & Khammash (2006). Finite State Projection. *J. Chem. Phys.*, 124:044104.
- Kemeny & Snell (1960). *Finite Markov Chains*. Van Nostrand.
- Briggs, Henson, McCormick (2000). *A Multigrid Tutorial*, 2nd ed. SIAM.
- Dendukuri, Chandrasekaran, Petzold (2025). Adaptive multigrid FSP for the RDME. *In preparation*.